

Summary

- 8+ years building AI/ML systems at scale, from model serving and optimization to large-scale training infrastructure.
- Shipped LLM serving, AI agent tooling, and ML training infrastructure across Kubernetes and cloud environments.
- Open source contributor: Apache Airflow @task API (shipped in 2.0), TFX, Triton Server.

Experience

- **Block Inc** *L6 Machine Learning Engineer, September 2024 - Present*
 - Built Kubernetes ML training platform on AWS serving 75+ monthly active users across Block.
 - Maintained Gondola, Block's unified ML inference service, serving 200K+ model requests per second.
 - Built inference logging system processing 19B+ records monthly at 7.5K rows/sec.
 - Built and maintained Slurm ML training cluster on GCP for 2+ years, enabling a NeurIPS paper.
 - Added HTTP MCP support to our model hosting service for automatic model development and debugging.
 - Drove platform strategy across Cloud Economics, AWS, and 5+ internal ML teams.
- **Block Inc** *L5 Machine Learning Engineer, March 2023 - September 2024*
 - Co-authored multi-model hosting endpoints reducing inference cost by 50%.
 - Co-authored Python SDK for automatic model deployment, reducing deploy times to hours instead of days.
 - Designed self-hosted LLM serving gateway with OpenAI-compatible API and vLLM within Block.
 - Presented in industry conference about scaling up Gondola, our unified ML model hosting at Block.
- **Twitter/X** *Senior Machine Learning Engineer, April 2022 - February 2023*
 - Co-authored and launched X Inc's large-scale ML training cluster using Slurm and Ansible.
 - Led team of 8 engineers defining roadmap, execution cadence, and technical direction for ML Pipelines.
 - Partnered with Spotify and Google to define reusable component standards for ML retraining pipelines.
- **Twitter** *Machine Learning Engineer II, October 2020 - April 2022*
 - Led adoption and migration of production model retraining to Kubeflow, standardizing pipeline scheduling.
 - Led development of automatic CI/CD system for ML pipelines.
 - Launched Ads Click model retraining in GCP with Ads team, accelerating model iteration cycles.
- **Twitter** *Machine Learning Engineer, September 2019 - October 2020*
 - Developed hyperparameter optimization tooling (Tuner) used in RecSys2020 paper, increasing Twitter Who To Follow follow rate by 15%.
- **Project EPIC, University of Colorado Boulder** *Research Assistant, February 2017 - August 2019*
 - Analyzed 2000 tweets/minute of real time data by building Big Data pipeline using Spark, Kafka and Cassandra.
 - Deployed system prototype with Kubernetes using custom microservices written in Go and Python.
- **Twitter** *Software Engineering Intern, May 2018 - August 2018*
 - Developed CLI to deploy Apache Airflow instances internally.
 - Helped reduce ML model training and deploy time by 1/4 developing ML Workflows.
- **inLab FIB** *Web Development Intern, March 2015 - January 2017*
 - Built REST API for faculty data - Average usage of 25.000 daily accesses. (Django, Python)

Education

- **University of Colorado Boulder** Boulder, CO
M.S., Computer Science (GPA 3.97) *August 2017 - August 2019*
 - **Courses:** Analysis and Design of Algorithms, User Centered Design, Database Systems, Big Data Architecture, Machine Learning, Computational Genomics, Natural Language Processing.
- **Universitat Politècnica de Catalunya** Barcelona, Spain
B.S., Computer Science (8.8/10) *September 2013 - July 2017*
 - **Courses:** Data Mining, Data Compression, Programming Languages, Software architecture, Databases, Parallelism, Project Management, Web architecture, Android, NoSQL databases, Reflective programming.

Volunteering

- **TFX Addons SIG** *Core Contributor, May 2021 - March 2023*
 - Release manager from version 0.1.0 to date.
 - Contributor and maintainer for project CI/CD system, XGBoostEvaluator, FeastExampleGen and others.
 - Collaborator for ModelCardTooling migration mentoring Twitter team members to contribute to project.
- **AIP-31: Apache Airflow @task API** *Core Contributor, January 2020 - July 2020*
 - Proposed and implemented improvement for Airflow DAG user experience adding explicit XCom data.
 - Released in Airflow 2.0.0 and used in one of the main tutorials for the project.
- **HackCU, University of Colorado Boulder** *Tech director, February 2017 - April 2019*
 - Co-led Colorado's biggest student hackathon with 600 attendees and a 15 people core team.
- **HackUPC, Universitat Politècnica de Catalunya** *Co-Director, November 2015 - September 2018*
 - Co-led Spain's biggest student hackathon with 500 attendees and a 20 people core team.
 - Led web development team with 5 people for 2 years, developing the event website and registration platform.

Awards

- **Google OSS Peer Bonus for contributions to TFX and TFX Addons** (2022)
- **Grant winner at MD5 Physical Cyber Hackathon.** (2018)
- **Balsells Fellowship.** (2017-2019, two years of funding for graduate studies)
- **Balsells Mobility program.** (2017, five months of funding for research)

Publications & Acknowledgments

- Jennings A., Gerard C., Kenneth M. A., Leysia P., and Rebecca M. **Incorporating Context and Location Into Social Media Analysis: A Scalable, Cloud-Based Approach for More Powerful Data Science.** *HICSS 52 (accepted).*
- Gerard C. **Social Media Analysis for Crisis Informatics in the Cloud** *MS Thesis*
- Chamberlain B., Rossi E., Shiebler D., Sedhain S., Bronstein M. **Tuning Word2vec for Large Scale Recommendation Systems.** *RecSys 2020.* (acknowledged for hyperparameter optimization tooling)

Skills

- **ML/AI:** PyTorch, TensorRT, Triton, vLLM, ONNX, LLM serving, model optimization. **Infrastructure:** Kubernetes, EKS, Helm, ArgoCD, Terraform, AWS, GCP. **MLOps:** Kubeflow Pipelines, Apache Airflow, Spark, Kafka. **Languages:** Python, Go.
- **Native:** Catalan, Spanish. **Advanced:** English.
- **Hobbies:** Skiing, Running, Photography, Roller hockey, Piano, Swim.